

The Capital Asset Pricing Model

In finance, the **Capital Asset Pricing Model (CAPM)** is a model that is used to determine the rate of return of any assets; it helps us determine whether we can add the assets to create a diversified portfolio. This model usually divides the assets based on sensitivity and risk, which is represented by β in the financial industry. It also considers the expected returns of the market and theoretical risk-free assets.

We will cover the following topics in this chapter:

- Understanding the CAPM
- Understanding and calculating the beta of a security
- Calculating the beta of a stock
- The CAPM formula
- Calculating the expected return of stock
- Understanding the Sharpe ratio

Technical requirements

In this chapter, we will be using Jupyter Notebook for coding purposes. We will be using the pandas, NumPy, matplotlib, and SciPy libraries.

Understanding the CAPM

In 1960, a meeting was held between Harry Markowitz and William Sharpe. Markowitz offered the 26-year-old Sharpe an opportunity to continue his research on how various stocks functioned together. Sharpe wrote his thesis on this subject and created one of the most important models in financial history.

The premise of CAPM is not that different to the one created by Markowitz. Investors are risk-averse. They prefer earning a higher return but are also cautious about the risks they might face and want to

optimize their portfolios in terms of both risk and return. Investors are unwilling to buy anything other than an optimal portfolio that optimizes their expected returns and standard deviation.

William Sharpe introduced the concept of a market portfolio as a bundle of all possible investments in the world, both bonds and stocks. The risk-return combination of this portfolio is superior to that of any other portfolio. The expected return of the market portfolio coincides with the expected return of the market and, because this is a diversified portfolio, it makes sense that it is optimal in terms of risk. It contains no idiosyncratic risk. The only risk faced by investors who own the market portfolio is systematic risk. We would expect this portfolio to lie somewhere on the efficient frontier as it is an efficient portfolio – the most efficient, in fact, of all portfolios.

Sharpe's CAPM contains the following:

- **Market portfolio:** A bundle of all possible investments in the world (both bonds and stocks). The risk-return combination of this portfolio is superior that of any other portfolio.
- **Risk-free asset:** An investment with no risk (zero standard deviation). It has a positive rate of return, but zero risk associated with it.
- **Beta coefficient:** This helps us to quantify the relationship between a security and the overall market portfolio.
- **Capital market line:** This is the line that connects the risk-free rate and is a tangent to the efficient frontier. The point at which the capital market line is a tangent to the efficient frontier is the market portfolio.

This looks like a simple extension of the ideas introduced by Markowitz. However, the CAMP assumes the existence of a risk-free asset and an

investment with no risk and zero standard deviation. It has a positive rate of return but zero risk associated with it. There are people who are so risk-averse that they prefer to be 100% certain that their investment has no risk. These people will be willing to accept a lower rate of return. The market portfolio has a lower expected rate of return because, in an efficient market, investors are compensated for the additional risks they are willing to bear once they own an efficient portfolio. They usually can't arbitrage the system and earn a higher expected return for the same level of risk. This means that a higher rate of returns means more risk and a lower level of risk means lower returns. If we have no risk, this translates to the lowest level of expected return, which is why we can expect the risk-free rate to provide a low level of expected returns. Rational investors will form their portfolios by considering both the risk-free stocks and how much they are going to invest in the market portfolio. This question really depends on how much they want to earn. Every investor can invest in a combination of risk-free assets and the market portfolio, according to their particular appetite for risk. If they are willing to risk more, they will hold a greater portion of the market portfolio. If they are unwilling to risk their money, they will buy more of the risk-free assets and less of the market portfolio. Investors who are interested in high expected returns will be able to borrow money and invest it in the market portfolio, going even further along the capital market line.

In the next section, we will discuss the relationship between the securities of individuals and the market portfolio. This will move us one step closer to learning how to build expectations about the prices of real-world assets.

The beta of securities

In this section, we will introduce the concept of beta, which is one of the main pillars of the CAPM. Beta helps us to quantify the relationship between the security and the overall market portfolio. Remember the market portfolio is made up of all the securities in the market: securities with low expected returns and securities with high expected returns. If there is an economic crisis, it is reasonable to expect the prices of most assets that make up the market portfolio to decrease and the market portfolio to experience a negative rate of return of, for example, -5%. In this case, investors can't protect themselves through diversification, because this is a systemic risk.

However, some securities in the market portfolio are less risky. They have a lower standard deviation and will decrease in value less than the market standard. Let's imagine that we have stock *A* and stock *B*. Stock *A* is less volatile than the market portfolio and so we only lose 3%. Stock *A* is an example of a security in the market portfolio that is less risky in times of volatility. The value of stock *B*, however, decreases by 7%, which is more than stock *A*. We can therefore consider stock *B* to be more volatile than stock *A*. Since stock *B* is volatile, when the economy recovers from the crisis, the market portfolio will do well and stock *B* may earn us a higher rate of return than stock *A*.

This example allows us to see that stocks might have a different behavior regarding their overall market performance. Some stocks are safer and will lose less and earn less than the market portfolio, while other stocks are riskier and will do well when the economy flourishes, but will perform poorly in times of crisis. This is precisely where beta (β) comes in handy. Beta allows us to measure the relationship between a stock and the market portfolio. It can be calculated as the covariance between the stock and the market divided by the variance of the market:

$$\beta_p = \frac{Cov(r_p, r_b)}{Var(r_b)}$$

The preceding formula measures the market risk, which cannot be avoided through diversification. The riskier a stock, the higher its beta. Let's look at a few important points that will help us understand beta:

- A beta of 0 means the stock has no relationship to the market.
- Betas that are below 1 are called defensive, because if the market does poorly, these stocks will typically lose less.
- Stocks with a beta of 1 will perform in the same way as the market.
- Stocks with beta of more than 1 are riskier than the market. They may do better than the market when the economy flourishes and lose more when it goes down.

We can understand the concept of beta by considering real-life examples. Some companies, such as Walmart, are less dependent on the economic cycle. Their clients must buy food and household products. In times of crisis, they may buy less food, but they can't give up food completely. An automobile company, however, is more dependent. During a recession, people are likely to prefer to use their old vehicles instead of buying a new vehicle. Car makers can therefore expect a significant slump in their revenues.

These examples basically illustrate how a recession might have a different impact on different businesses. The beta of these companies indicates whether they are riskier than the market.

Calculating the beta of a stock

In this section, we will see how we can find the beta value of a stock. The beta is typically measured using data from the past five years. We will be using data from Microsoft and the S&P 500 for the period between January 1, 2012 and December 31, 2016. We will calculate the beta of Microsoft and we will approximate the development of the market with the S&P 500.

Initially, we will import all the libraries that are required, such as [numpy](#) and [pandas](#). We import the dataset of the Microsoft and S&P 500 as follows. The S&P index is given by the `^GSPC`. The S&P 500 is a capitalization-weighted index, and is associated with many ticker symbols, such as `^GSPC`, `INX`, and `$SPX`, depending on market or website. The S&P 500 differs from the Dow Jones Industrial Average and the NASDAQ Composite index because of its diverse constituency and weighting methodology.

```
In [1]:
import numpy as np
import pandas as pd
data = pd.read_csv('MSFT_S&P.csv', index_col = 'Date')
data
```

The preceding code displays the following output:

Date	MSFT	^GSPC
2012-01-03	23.031868	1277.060059
2012-01-04	23.573898	1277.300049
2012-01-05	23.814798	1281.060059
2012-01-06	24.184755	1277.810059
2012-01-09	23.866419	1280.699951
2012-01-10	23.952452	1292.079956
2012-01-11	23.849216	1292.479980
2012-01-12	24.090115	1295.500000
2012-01-13	24.305206	1289.089966
2012-01-17	24.313807	1293.670044

We need two values: the covariance between Microsoft and the S&P 500, and the variance of the S&P 500. We start by calculating the logarithmic returns and storing them in a `sec_returns` variable, as follows:

```
In [2]:
sec_returns = np.log( data / data.shift(1) )
```

Once we've done that, we can create a covariance matrix between MSFT and the market, as follows:

```
In [3]:
cov = sec_returns.cov() * 250
cov
```

The output of the preceding code is as follows:

	MSFT	^GSPC
MSFT	0.053781	0.018208
^GSPC	0.018208	0.016361

The `iloc` method allows us to obtain the covariance between the MSFT and S&P market as a float, as follows:

```
In [4]:
cov_with_market = cov.iloc[0,1]
cov_with_market
```

```
Out[4]:
0.01820843191039893
```

The preceding output is the nominator required to calculate the beta value, which is equal to the variance of the market. In our case, we can use a new variable called `market_var`, which contains the annual variance of the S&P 500:

```
In [5]:  
  
market_var = sec_returns['^GSPC'].var() * 250  
market_var
```

```
Out[5]:  
  
0.016360592699269063
```

Finally, we calculate the beta value of these stocks by using the formula discussed in the previous section:

```
In [6]:  
  
MSFT_beta = cov_with_market / market_var  
MSFT_beta
```

```
Out[6]:  
  
1.112944515219942
```

The beta value that we get is greater than one. This means that the MSFT stock is riskier than the market. It might do better than the market when the economy flourishes and lose more when it goes down. The next step is to verify whether our calculations are correct. We are currently using data from Yahoo Finance. Let's now go to Yahoo Finance to check whether the beta value we obtained is credible. We can allow ourselves a 2 - 3% difference, because the data and the estimation

methods may differ slightly, but the difference shouldn't be any bigger than that.

Once we go the Yahoo finance website, search for the MSFT stock and check the quoted beta coefficient, as shown in the following screenshot:

Valuation Measures		Trading Information	
Market Cap (intraday) ⁵	526.88B	Stock Price History	
Enterprise Value ³	782.77B	Beta (3Y Monthly)	1.09
Trailing P/E	50.57	52-Week Change ²	27.14%
Forward P/E ¹	21.37	S&P500 52-Week Change ³	5.55%
PEG Ratio (5 yr expected) ¹	1.77	52 Week High ³	116.18
Price/Sales (ttm)	7.49	52 Week Low ³	80.70

We can see that the beta value calculated for three years by Yahoo Finance is 1.09, which is similar to the value we obtained. This validates the beta coefficient that we calculated.

The CAPM formula

In the previous sections, we imagined that we were living in a world where all investors are rational, risk-averse, and willing to optimize their investment portfolios. We introduced the reader to the concept of a market portfolio and a risk-free asset. We also stated that investors make their decisions based on their risk appetite: those seeking higher expected returns will allocate a greater portion of their money to the market portfolio and less to the risk-free assets. Finally, we introduced the concept of beta, which measures how securities are expected to perform with respect to the entire market. We are now going to introduce the capital asset pricing model. The formula of the capital asset pricing model is as follows:

$$r_i = r_f + \beta_{im}(r_m - r_f)$$

The CAPM formula has the following important parameters:

- **r_f (Risk-free)**: The risk-free rate of return. This is the minimum amount of compensation an investor would expect from an investment.
- **β_{im} (Beta coefficient)**: The beta value between the stock and the market. This coefficient specifies how risky a specific asset is with respect to the market.

The CAPM formula has the following important parameters:

- **r_f (Risk-free)**: The risk-free rate of return. This is the minimum amount of compensation an investor would expect from an investment.
- **β_{im} (Beta coefficient)**: The beta value between the stock and the market. This coefficient specifies how risky a specific asset is with respect to the market.
- **r_m (Market risk premium)**: The market return. The amount of compensation an investor would expect when buying the market portfolio.

Let's take a look how the CAPM formula works. We start with a risk-free rate of return (r_f), which is the bare minimum an investor would accept in order to buy a security. Since the investor must be compensated for the risk they're taking, we need the equity premium component given by the expected return of the market portfolio minus the risk-free rate. It is reasonable to expect this incentive since the security has a risk attached to it. This is the most widely used model by practitioners.

Calculating the expected return of a stock (using the CAPM)

We have covered a lot of concepts related to the CAPM. We have also discussed the beta coefficient and the CAPM formula. Let's now look at how we can compute the expected returns of a stock using the CAPM. First of all, we need to calculate the beta coefficient before we can apply it to the CAPM formula. Here, we will use Python and implement all the formulas of CAPM using Python. We will be using the same dataset that we used to calculate the beta coefficient and follow the same process as before:

```
In [2]:

import numpy as np
import pandas as pd

data = pd.read_csv('MSFT_S&P.csv', index_col = 'Date')
data

sec_returns = np.log( data / data.shift(1) )
cov = sec_returns.cov() * 250
cov_with_market = cov.iloc[0,1]
market_var = sec_returns['^GSPC'].var() * 250

MSFT_beta = cov_with_market / market_var
```

From the preceding code, we get the beta value, which is stored in the MSFT_beta variable.

Let's take another look at the CAPM formula:

$$r_i = r_f + \beta_{im}(r_m - r_f)$$

We can get an approximation of the risk-free trade value from Bloomberg's website. Assuming a risk-free rate of 2.5% and a risk premium of 5%, we now need to estimate the expected return of the Microsoft (MSFT) stock. We will use the CAPM formula and create an MSFT_er variable, which gives the expected return of the MSFT stock, as follows:

```
In [3]:  
  
MSFT_er = 0.025 + MSFT_beta * 0.05  
MSFT_er  
  
Out[3]:  
  
0.0806472257609971
```

MSFT_er variable, which gives the expected return of the MSFT stock, as follows:

```
In [3]:  
  
MSFT_er = 0.025 + MSFT_beta * 0.05  
MSFT_er  
  
Out[3]:  
  
0.0806472257609971
```

The value we get is 8.06%. This is the return on the investment we might expect when buying the MSFT stock. This technique can be applied for any listed company that you are interested in.

Applying the Sharpe ratio in practice

Now that we know how to calculate a stock's return and assess its risk through its variance and standard deviation, we're ready to take another look at the Sharpe ratio. We discussed this topic in the previous chapter, but we will go into more detail in this section. As mentioned earlier, rational investors want to maximize their returns and are risk averse, which means they want to minimize the risk they face and invest in less volatile securities. They want less uncertainty and more clarity about an investment's rate of return. Rational investors are afraid that a

risky investment would cause significant losses and want to invest their money in securities that are less volatile. It becomes obvious that the two dimensions must be combined.

We need a measure of the risk-adjusted return, a tool that will allow us to compare different securities. This is where the Sharpe ratio comes in. It is a great way to make comparisons between stocks and portfolios and decide which one is preferable in terms of risk and return.

The Sharpe ratio is given by the following formula:

$$S = \left(\frac{R_p - R_f}{\sigma_p} \right)$$

Here, R_p is the expected portfolio return of the stock, p ; R_f is the risk-free return; and σ_p is the portfolio standard deviation of the stock, p .

In the numerator of the preceding formula, we have the excess return of a stock. We have to subtract the risk-free rate (R_f) from the expected portfolio return (R_p). In the denominator, we have the standard deviation of the stock. If we increase the stock's expected rate of return, its Sharpe ratio becomes higher. Logically, if we increase the stock's standard deviation, its Sharpe ratio becomes lower.

This helps us to compare financial securities and portfolios quickly. It also allows us to understand whether a certain investment fund is performing in a satisfactory manner on a risk-adjusted basis. However, we should be aware that this often comes at the expense of a riskier portfolio.

Obtaining the Sharpe ratio in Python

We will now continue our analysis of the Microsoft stock with regards to the S&P 500, which will represent the market. We will create a variable named Sharpe, and this variable will be assigned with the following formula:

$$\underline{\text{sharpe}} = \frac{\overline{r_{msft}} - r_f}{\sigma_{msft}}$$

We have to subtract the risk-free rate from the expected rate of return of the stock. We calculated the expected returns in a previous section and got a `MSFT_er` value of 0.0806. We can use the `MSFT_er` variable directly as the risk-free rate of Microsoft (r_{msft}). We also assumed the risk-free rate (r_f) to be 2.5%, which is equal to 0.025. To calculate the standard deviation of MSFT stock, we will use the `std()` method to obtain the volatility and annualize it by multiplying it by the square root of 250. The complete code to calculate the Sharpe ratio can be written as follows:

```
In [2]:  
  
Sharpe = (MSFT_er - 0.025) / (sec_returns['MSFT'].std() * 250 ** 0.5)  
Sharpe  
  
Out[2]:  
  
0.23995502423552612
```

The output is approximately 24%. We can use this ratio when we want to compare different stocks in stock portfolios.

Measuring alpha and verifying the performance of a portfolio manager

So far, we have learned about regression, the Markowitz efficient frontier of investment portfolios, and William Sharpe's capital asset pricing model. We have also learned how to calculate a portfolio's risk,

return variance, and covariance. We know what beta is and how it can be interpreted in the context of Sharpe's capital market line.

In this section, we'll add another tool to our arsenal of financial knowledge. We'll learn how to interpret the intercept of the CAPM model, **alpha** (α). In the world of finance and investments, alpha is often thought of as a measure of the performance of a fund manager. The standard CAPM assumes an efficient financial market and an alpha of 0.

The following is the formula for the CAPM:

$$r_i = \alpha + r_f + \beta_{im}(r_m - r_f)$$

Given that the beta (β_{im}) multiplied by the equity risk ($r_m - r_f$) premium gives us the compensation for the risk that's been taken with the investment, alpha shows us how much we get without bearing the extra risk. A great portfolio manager is someone who outperforms the market and can achieve a high alpha. Conversely, a poor investment professional may obtain a negative alpha, meaning they underperformed with respect to the market.

In order to outperform the market, a portfolio manager can't simply invest in a diversified portfolio and sit back and relax. They have to select the winning portfolio of stocks. This is easier said than done, but investment professionals have a profound knowledge of certain industries, companies, and the stock market. They hand-pick successful investments and use active trading strategies. The term **active trading** means that an investor will try to take advantage of short-term price changes. We don't simply buy a stock portfolio that replicates the S&P 500.

We have to pick a portfolio of stocks that will win. Because other investments will eventually catch up and the price of these shares will normalize, we will have to sell the shares and select a new group of winning companies. This is active trading. The typical time period of active investments depends on the situation but ranges from a day to several months. Active trading is the basis on which investment professionals justify their fees.

To recap, there are two different types of trading:

- **Passive trading:** This trading technique involves investing in a market index (the S&P 500) and waiting patiently.
- **Active trading:** This trading involves adjusting investment

portfolios on a frequent basis, based on their growth.

If an investment professional charges you 1% of the invested amount, they need to outperform the market by over 1%, otherwise you'd be better off investing in a passive fund that charges you a tiny portion of the invested income. Don't forget that the alpha is only comparable when the risk profile of the investments being compared is similar. We shouldn't compare a portfolio of small illiquid companies operating in an emerging market with a well-diversified portfolio of bluechip companies based in the US. Many investment managers struggle to produce an alpha value that is consistently over zero. Many investors are aware that well-diversified and passive index-tracking funds that charge low fees are very efficient, and are starting to consider these as a solid alternative to active management strategies.

William Sharpe believes that investors should bet on the efficiency of markets and shouldn't bother paying high management fees, given that only 2% funds have a positive alpha that is consistently different than zero.

Let's recap the ideas presented in this section.

The alpha is given by the following formula:

$$r_i = \alpha + r_f + \beta_i(r_m - r_f)$$

In the world of finance and investments, alpha is often thought of as a measure of the performance of a fund manager has been. The standard CAPM assumes an efficient financial market and an alpha of 0. Given that the beta multiplied by the equity risk premium gives us the compensation for risk that's been taken with the investment, the alpha shows us how much return we get without bearing extra risk. A great portfolio manager, someone who outperforms the market, can achieve a high alpha. Conversely, a poor investment professional may obtain a negative alpha, meaning they underperformed the market.

There are various different types of investment strategy:

Type of correlation	Description
Passive investing	This consists of buying a portfolio of assets and holding it for a long time regardless of the short-term macroeconomic developments.
Active investing	This refers to frequent trading based on the expectations of macroeconomic and company-specific developments.
Arbitrage trading	This is where we find pricing discrepancies on the market and exploit them in order to make a profit without assuming additional risk.
Value investing	This is where we invest in specific companies, in the hope that they will outperform their peers.

Different use cases of the CAPM using the scipy library

In this section, we will take a look at how we can implement the CAPM using the scipy library.

We will begin importing the scipy library, as follows:

```
In [2]:  
  
from scipy import stats
```

Then, we will import pandas and pandas_datareader, which will be used to pull the dataset from the API, as follows:

```
In [17]:
```

```
import pandas as pd
```

```
In [13]:
```

```
import pandas_datareader as web
```

After that, we import the data with the help of pandas_datareader. The index we have provided is basically the SPY, which is an ETF. In this context, index basically means a market:

```
In [14]:
```

```
spy_etf = web.DataReader('SPY','google')
```

We use the info function to see the details of the columns extracted, as follows:

```
In [21]:
```

```
spy_etf.info()
```

```
Out[21]:
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
DatetimeIndex: 1897 entries, 2010-01-04 to 2017-07-18
```

```
Data columns (total 5 columns):
```

```
Open      1879 non-null float64
```

```
High      1879 non-null float64
```

```
Low       1879 non-null float64
```

```
Close     1897 non-null float64
```

```
Volume    1897 non-null int64
```

```
dtypes: float64(4), int64(1)
```

```
memory usage: 88.9 KB
```

We can also see the first five rows of the dataset retrieved from the API using the head() function:

```
In [22]:
```

```
spy_etf.head()
```

The preceding code generates the following output:

Date	Open	High	Low	Close	Volume
2010-01-04	112.37	113.39	111.51	113.33	118944541
2010-01-05	113.26	113.68	112.85	113.63	111579866
2010-01-06	113.52	113.99	113.43	113.71	116074402
2010-01-07	113.50	114.33	113.18	114.19	131091048
2010-01-08	113.89	114.62	113.66	114.57	126402764

If you take a look at SPY ETF, we have about 2,000 entries, each of which have open, high, low, and close columns. These results are actually indicators, rather than the actual S&P 500. SPY ETF is an exchange-traded fund that represents the S&P 500.

Let's now create a start date and an end date:

```
In [18]:
```

```
start = pd.to_datetime('2010-01-04')  
end = pd.to_datetime('2017-07-18')
```

Now, let's say that our portfolio strategy is to invest entirely in Apple stocks. We will use `pandas_datareader` and retrieve the Apple stock with the preceding start and end dates from Google Finance, as follows:

```
In [24]:
```

```
aapl = web.DataReader('AAPL','google',start,end)
```

Once we have executed the preceding statement, we can check out the head of the dataframe, as follows:

```
In [27]:
```

```
aapl.head()
```

The preceding code generates the following output:

Date	Open	High	Low	Close	Volume
2010-01-04	30.49	30.64	30.34	30.57	123432050
2010-01-05	30.66	30.80	30.46	30.63	150476004
2010-01-06	30.63	30.75	30.11	30.14	138039594
2010-01-07	30.25	30.29	29.86	30.08	119282324
2010-01-08	30.04	30.29	29.87	30.28	111969081

The CAPM basically states that there should be a relationship between the performance of Apple's stocks and the overall market performance. Here, the market data is the data that we extracted for SPY, which is an **exchange-traded fund (ETF)**. Let's visualize this data. Import the matplotlib library, as follows:

```
In [28]:
```

```
import matplotlib.pyplot as plt  
%matplotlib inline
```

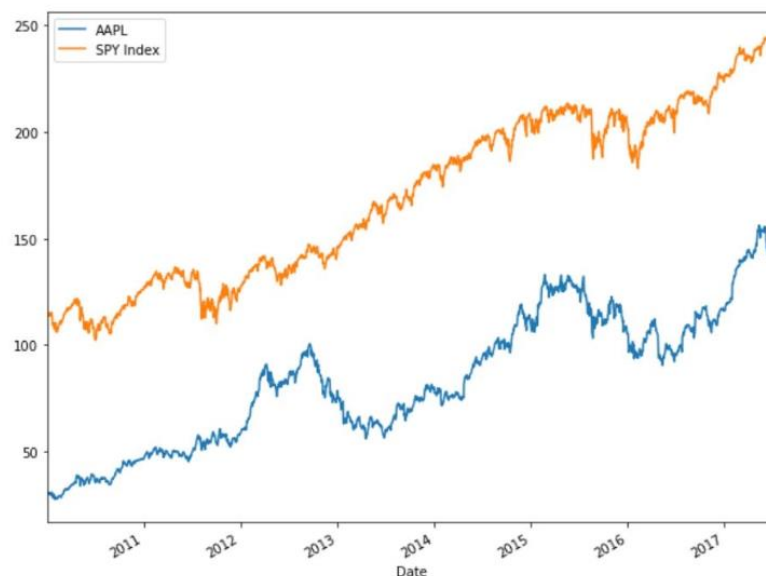
We will be plotting the Close column of the aapl dataframe along with the Close column of the spy_etf dataframe. We will consider a plot size of (8, 5), as follows:

```
In [29]:
```

```
aapl['Close'].plot(label='AAPL',figsize=(10,8))  
spy_etf['Close'].plot(label='SPY Index')  
plt.legend()
```

The output of the preceding code is shown in the following screenshot:

```
Out[29]: <matplotlib.legend.Legend at 0x1c4e9221cf8>
```



From the preceding output, we can see Apple and the index of the S&P market 500, or an exchange-traded fund, which essentially mimics its behavior. We now need to find the beta and the alpha, using the CAPM, so that we can compare Apple and the S&P ETF. To do this, we are going to return the cumulative returns.

The cumulative returns for Apple and the SPY ETF can be calculated as follows:

In [31]:

```
aapl['Cumulative'] = aapl['Close']/aapl['Close'].iloc[0]
spy_etf['Cumulative'] = spy_etf['Close']/spy_etf['Close'].iloc[0]
```

We basically divide all the Close values with the close price on the very first day. We create a new column called Cumulative in both the aapl and spy_etf dataframes. We now plot these cumulative values, as follows:

In [33]:

```
aapl['Cumulative'].plot(label='AAPL',figsize=(10,8))
spy_etf['Cumulative'].plot(label='SPY Index')
plt.legend()
plt.title('Cumulative Return')
```

The output of the preceding code is shown in the following screenshot:

Out[33]: <matplotlib.text.Text at 0x1c4e97460b8>



This output shows the cumulative return. If we invested \$1 in 2011, we can see that we could have got better returns if we had invested in Apple. Let's now check out the daily returns for these two stocks:

```
In [41]:
```

```
aapl['Daily Return'] = aapl['Close'].pct_change(1)
spy_etf['Daily Return'] = spy_etf['Close'].pct_change(1)
```

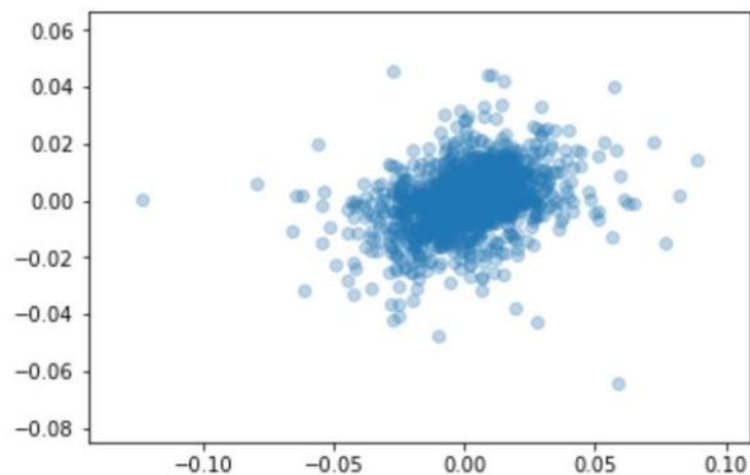
Here, we use the `pct_change()` function, where we provide one day as the parameter and create a new column called Daily Return for both `aapl` and `spy_etf`. Next, we will take the scatter plot and plot the daily return values of both `aapl` and `spy_etf` to check whether there is any correlation. Here, we also provide the value of `alpha`, which is equal to 0.3:

```
In [45]:
```

```
plt.scatter(aapl['Daily Return'],spy_etf['Daily Return'],alpha=0.3)
```

The output of the preceding code is shown in the following screenshot:

```
Out[45]: <matplotlib.collections.PathCollection at 0x1c4ea1ec518>
```



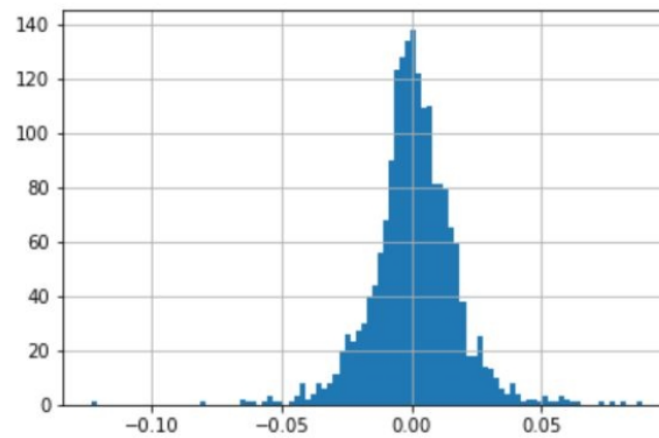
Form the preceding output, we can see that there is some correlation. We can also plot Daily Returns with the help of the histogram to see the highest frequency returns, as follows:

In [46]:

```
aapl['Daily Return'].hist(bins=100)
```

The output of the preceding code is shown in the following screenshot:

Out[46]: <matplotlib.axes._subplots.AxesSubplot at 0x1c4ea1a4908>



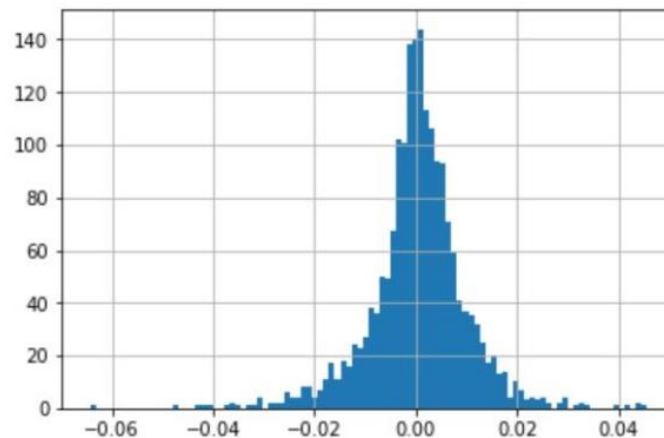
We can also take a look at the histogram of the spy_ETF Daily Return:

In [48]:

```
spy_ETF['Daily Return'].hist(bins=100)
```

The output of the preceding code is shown in the following screenshot:

Out[48]: <matplotlib.axes._subplots.AxesSubplot at 0x1c4e97c9cc0>



Let's calculate the beta and alpha values. The following code uses a tuple unpacking, which returns four different values. We are going to use the `stats.linregress()` method, as follows:

```
beta,alpha,r_value,p_value,std_err = stats.linregress(aapl['Daily Return'].iloc[1:],spy_etf['D
```

The following codes helps us to see the beta value:

```
In [38]:
```

```
beta
```

```
Out[38]:
```

```
0.19423150396392763
```

```
In [39]:
```

```
alpha
```

```
Out[39]:
```

```
0.00026461336993233316
```

```
In [40]:
```

```
r_value
```

```
Out [40]:
```

```
0.33143080741409325
```

Summary

In this chapter, we looked at the capital asset pricing model, which was used to determine the rate of return of an asset. We also discussed how to calculate the beta of the security, which helps us to determine which stock is riskier compared to the market. Then, we saw how to measure an alpha value and how to verify the performance of the portfolio. We also learned about the various use cases of the capital asset pricing model using the `scipy` library.

In the next chapter, we will discuss regression analysis in the financial domain.